

# EXAM PREPARATION AI AGENT

Rule-Based Intelligent Agent for Prioritizing Subject Study Time

## 1. Introduction / Explanation of the Agent

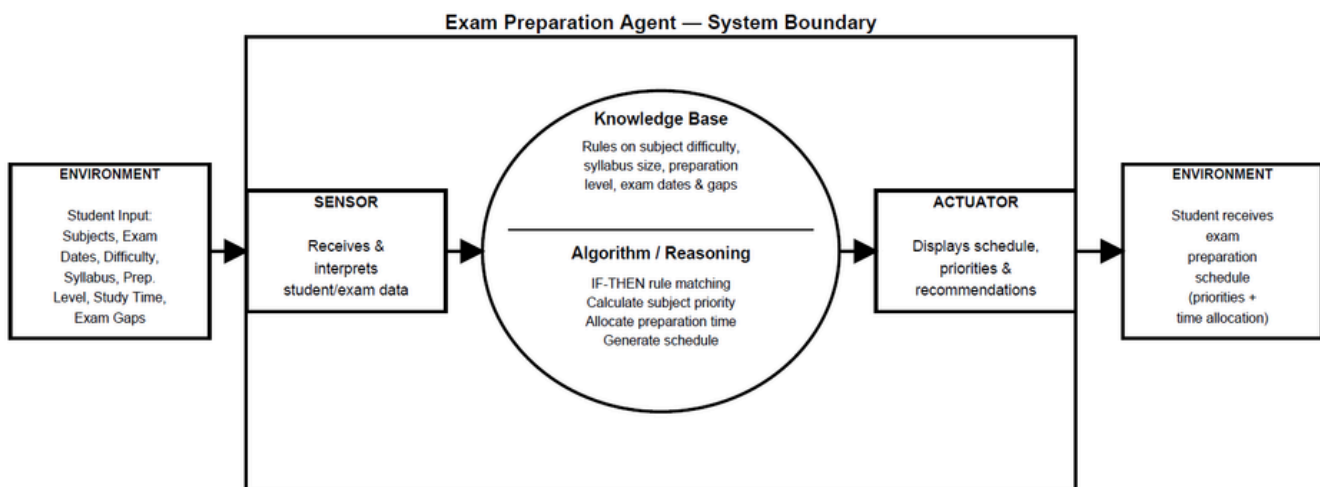
The **Exam Preparation AI Agent** is a software-based, rule-based intelligent agent that helps students plan how to prepare for multiple upcoming exams. Students commonly struggle to decide which subject to study first when several exams are scheduled close together. This agent solves that problem by automatically deciding which subjects deserve more attention and how the student's limited preparation time should be divided among them.

The agent receives information from the student such as the list of subjects, exam dates, subject difficulty, syllabus size, current preparation level, total available study time, and the gap between exams. It analyzes this information using a set of predefined logical (IF-THEN) rules — for example, a subject that is difficult, has a large syllabus, has low preparation, and an approaching exam date is automatically assigned a higher priority. Based on these rule evaluations, the agent decides how many study hours or days each subject should receive, and finally generates an exam preparation schedule that is displayed to the student.

Because the agent uses fixed, predefined rules rather than learning from data, it is classified as a **rule-based (simple reflex / model-based) AI agent** and does not use machine learning or deep learning.

**Example:** A student has Mathematics, Computer Science, and Science exams. Mathematics is difficult, has a large syllabus, the student's preparation is low, and the exam is approaching soon. The agent therefore assigns Mathematics the highest priority and allocates more preparation time to it, while subjects with lower difficulty or better existing preparation receive comparatively less time.

## 2. Graphical Representation of the Agent



### 3. Agent Formulation

| Component                 | Description                                                                                                                                                                                          |
|---------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Exam Preparation AI Agent |                                                                                                                                                                                                      |
| Agent                     | The student's academic/examination situation, consisting of: subjects, exam timetable, syllabus workload, the student's current preparation level, available study time, and the gaps between exams. |
| Environment               |                                                                                                                                                                                                      |

#### Environment Types

| Classification       | Explanation                                                                                                                                                                                   |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Partially Observable | The agent only sees the data the student provides (dates, difficulty, syllabus, etc.) and cannot observe the student's actual understanding, so it is partially observable.                   |
| Deterministic        | In this simplified rule-based version, the same inputs always produce the same priority and schedule, so it is treated as deterministic — although real student performance can be uncertain. |
| Sequential           | It is sequential because today's preparation decisions affect the student's readiness for later exams and future planning.                                                                    |
| Dynamic              | The environment is dynamic because exam dates keep approaching and the student's preparation level changes over time, even while the agent is planning.                                       |
| Discrete             | Inputs such as subjects, difficulty levels, and days remaining are treated as a finite, distinct set of values, making this simplified agent discrete.                                        |
| Single-Agent         | It is a single-agent system because only one student interacts with the agent to generate a personal study plan, with no competing or cooperating agents involved.                            |

#### Sensors / Percepts

Since this is a software agent, its "sensors" are digital inputs rather than physical hardware sensors. The agent perceives the following data:

- Exam dates
- Current date
- Subject difficulty
- Syllabus size
- Current preparation level
- Available study hours
- Gaps between exams

#### Actuators

The agent performs the following software actions:

- Generate a preparation schedule
- Assign preparation days to each subject
- Prioritize subjects
- Recommend revision periods
- Display reminders or recommendations

- Adjust the plan when input information changes

## Agent Goal

The agent's goal is to **maximize the student's exam readiness** by intelligently distributing limited preparation time among subjects based on difficulty, syllabus size, current preparation, and urgency.

## 4. PEAS Framework

| PEAS Component                 | Description                                                                                                                                                          |
|--------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>P</b> – Performance Measure | Exam readiness, appropriate time allocation, meeting all exam deadlines, prioritizing difficult/urgent subjects, balanced preparation, minimizing wasted study time. |
| <b>E</b> – Environment         | Student, subjects, syllabus, exam timetable, available study time, exam gaps.                                                                                        |
| <b>A</b> – Actuators           | Generate schedule, assign study days, prioritize subjects, recommend revision, send reminders.                                                                       |
| <b>S</b> – Sensors             | Exam dates, current date, difficulty, syllabus size, preparation level, available study hours.                                                                       |

The PEAS framework summarizes the agent design: **Performance measure** defines what the agent tries to achieve, **Environment** defines what it operates within, **Actuators** define what actions it takes, and **Sensors** define what input data it receives.